GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 02



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Recap of Previous Lecture



Topic Calendar

Topics to be Covered



Topic-1 Clock

[MCQ]

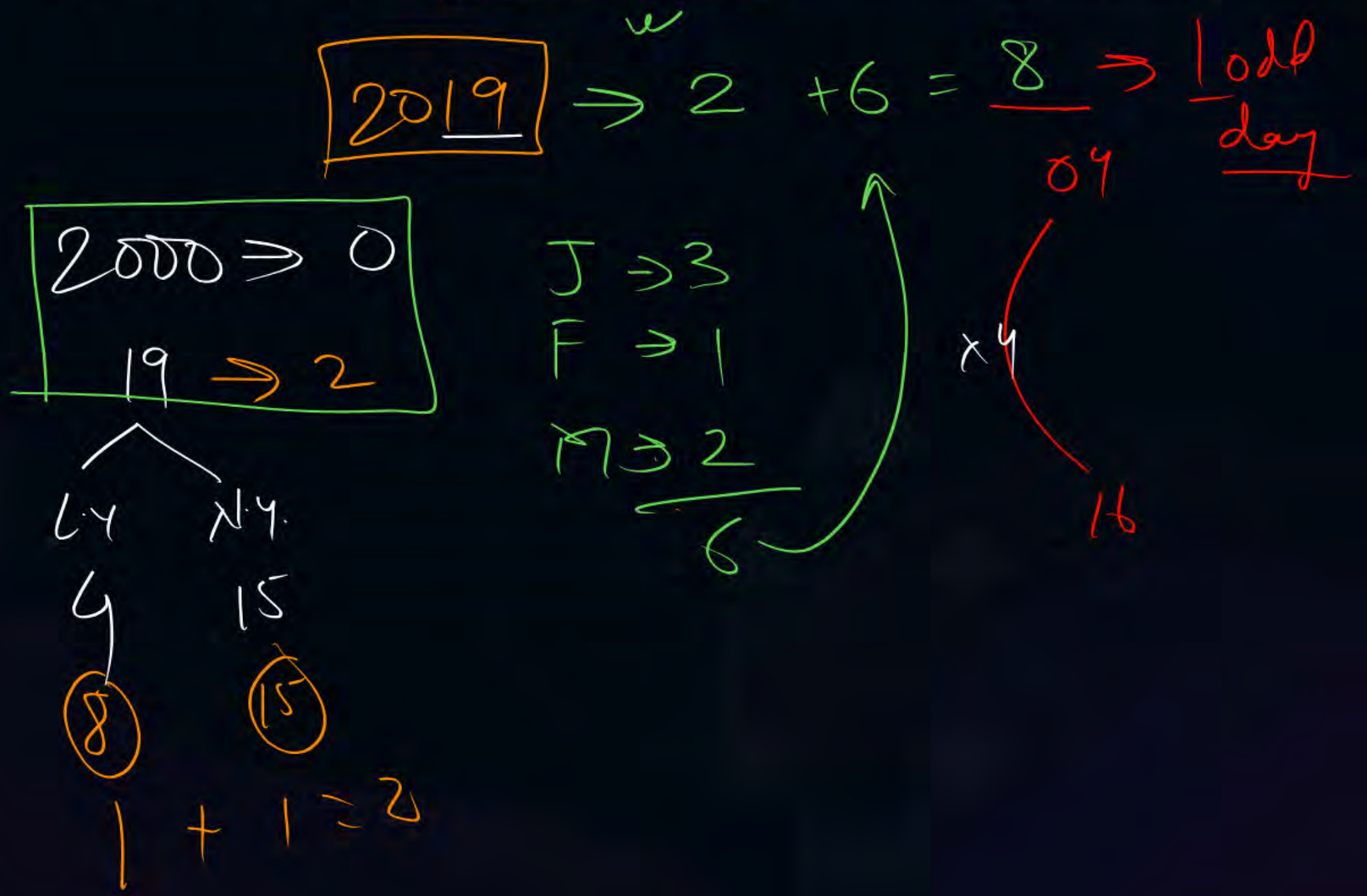
#Q. The GATE 2020 result was declared on March 16, 2020. What day of the week was it?

A. Saturday

B. Sunday

C. Monday

D. None



[MCQ]

Saturday

#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

A Monday

B Thursday

C Tuesday

D Saturday

1868 ⇒ 4 + 2 = 6

Assignment

1800 ⇒ 3

68 ⇒ 1

J ⇒ 3

F ⇒ 0 A ⇒ 3

17 ⇒ 3 S ⇒ 2

A ⇒ 2 O ⇒ 2

17 ⇒ 3

J ⇒ 2

J ⇒ 3

2

17 51

34 51

6 + 2 = 8

04

× 17

68

[MCQ]

#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

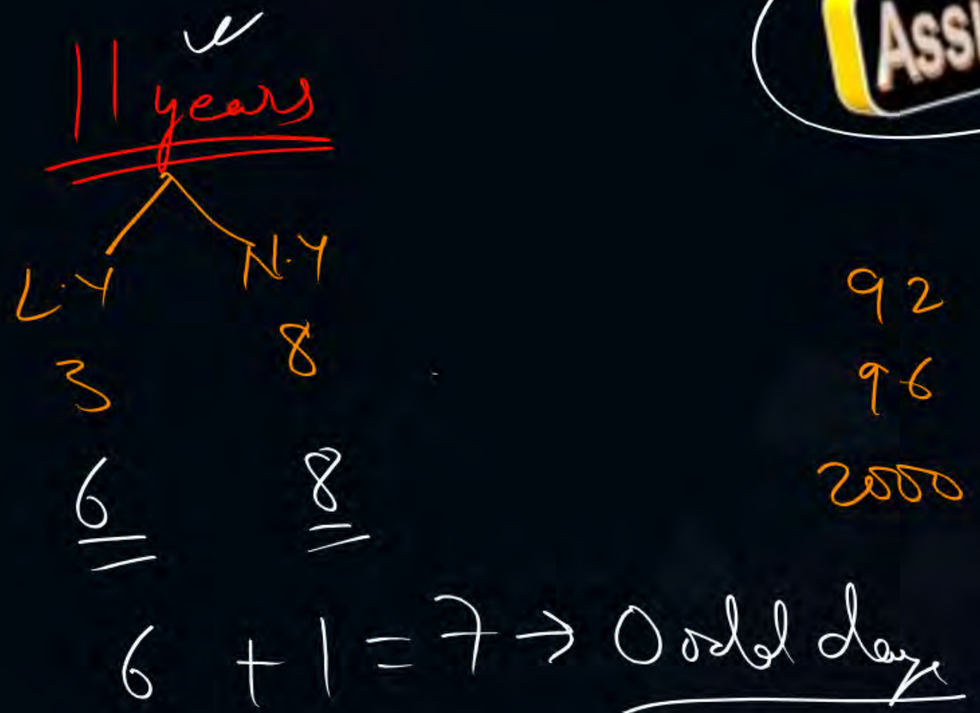


A Sunday

B Wednesday

C Thursday

D Friday



★ Date ⇒ Day ?

★

2011

CALENDAR REPETITION

2011 → 1

2012 → 2

2013 → 1

2014 → 1

2015 → 1

2016 → 2

2017 → 1

2018 → 1

2019 → 1

2020 → 2

2021 → 1

2022

[MCQ]

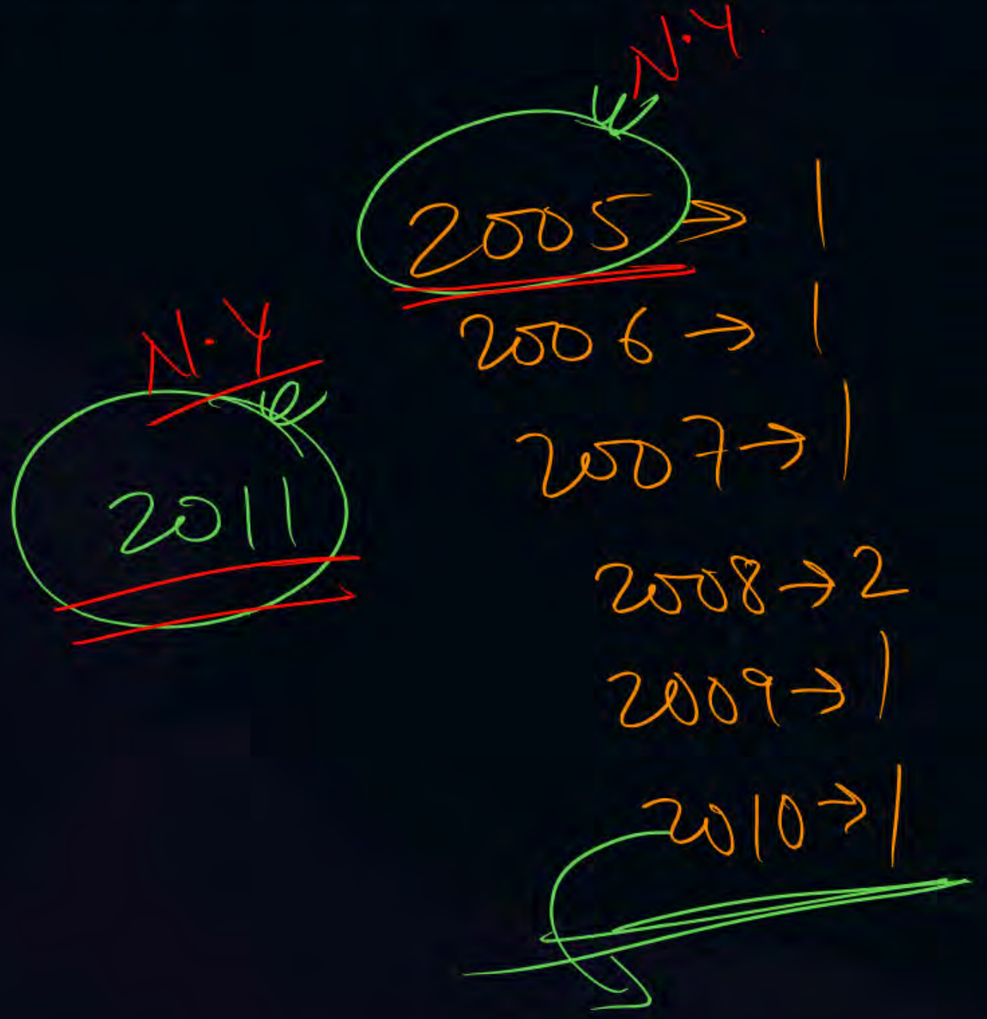
#Q. 2005 calendar is same as which of the given years?

A 2009

B 2016

C 2011

D 2015



[MCQ]

#Q. When does 2016 calendar repeats?

L.Y.

2016 + 28 = 2044

2005 + 6 = 2011

L.Y. ⇒ 28 yrs

N.Y. ⇒ 6 yrs

N.Y. ⇒ 11 yrs

N.Y. ⇒ 11 yrs

L.Y. ⇒ 28 yrs

A. 2027 ✗

B. 2040

2004 + 28 = 2032

2012 + 28 = 2040

C. 2021 ✗

D. 2044

2017 + 6 = 2023

Conditional Question

Q. If a year consist of 800 days and every week has 11 days, then how many odd days left particular year has?

11)800(72
77
30
22
8

Conditional Question

Q The last day of a century CANNOT be which day?

Tue | Thurs | Sat

100	5	Friday
200	3	Wed
300	1	Mon
400	0	Sun

[MCQ]

#Q. The day of the March 16th of any year is the same day of the week as the corresponding date in which month of the same year?

A. July

B. November

C. April

D. September

March ⇒ 3
April ⇒ 2
May ⇒ 3
June ⇒ 2
July ⇒ 3
Aug ⇒ 3

Sep ⇒ 2
Oct ⇒ 3
⇒ Nov

#Q. A person was born on fifth Monday of February in a particular year.
Which one of the following statements is correct based on the above information?.

A. The 2nd February of that year is a Tuesday.

B. There will be five Sundays in the month of February in that year.

C. The 1st February of that year is a Sunday.

D. All Mondays of February in that year have even dates.

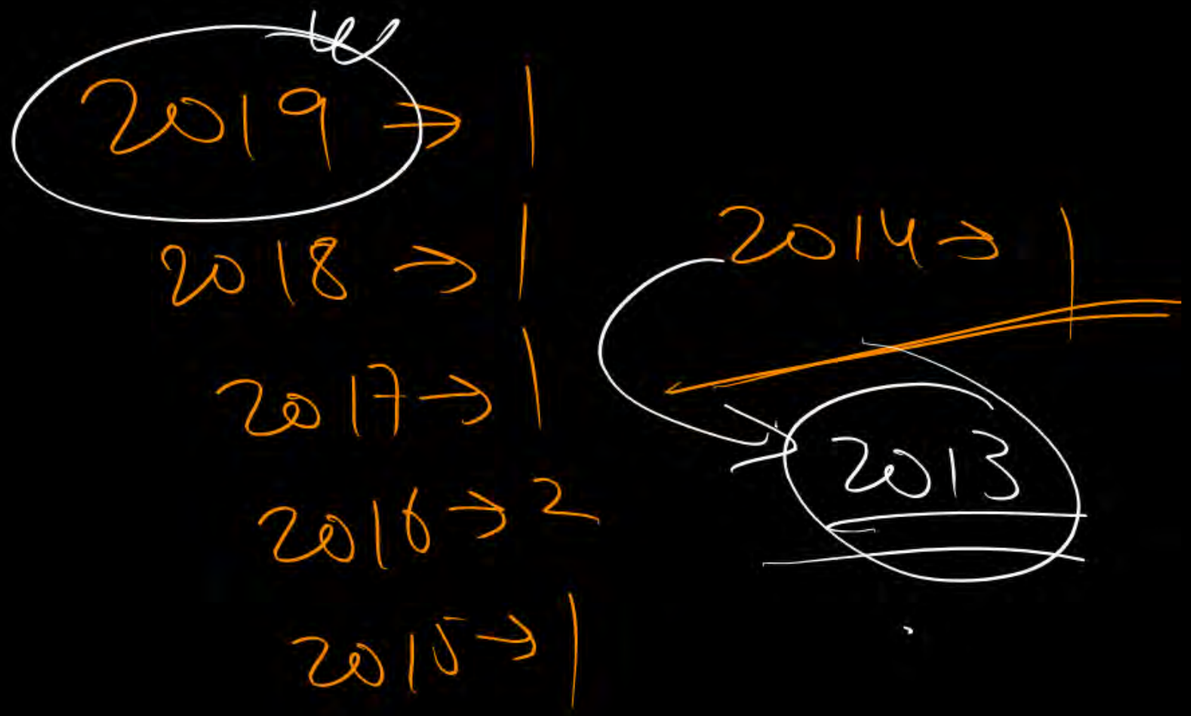
#Q. For the year 2019, which of the previous year's calendar can be used?

A 2011

B 2012

C 2013

D 2014



CLOCK



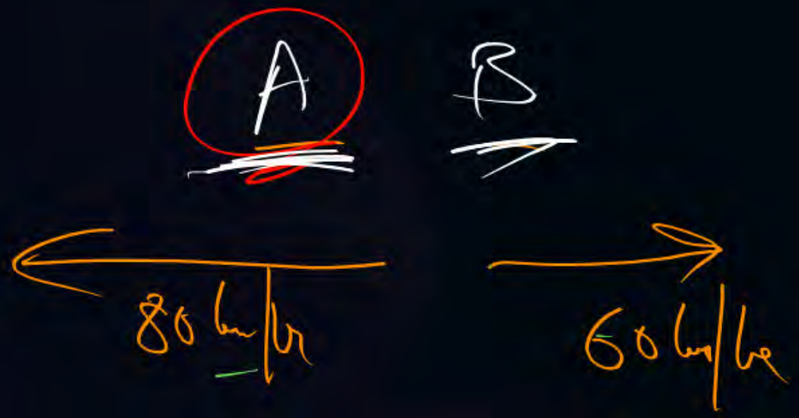
Relative Speed:

?

opp ⇒ (+)

Same ⇒ (−)





Relative Speed:

1. Hands — Minimum → 2

Hour hand (H·H)
Minute hand (M·H)

2. Movement — Same

3. Relative Speed (–)

Dial of Clock

$\frac{360^\circ}{12} = 30^\circ$



Dial of Clock

M.H. $= \frac{30^\circ}{5} =$ 6°/minute

H.H $= \frac{30^\circ}{60} = \frac{1}{2}$°/hour

$= 0.5$°/minute



To be Noted:

- ✓ Minute Hand Covers 30° in 5 minutes 30°/5 =

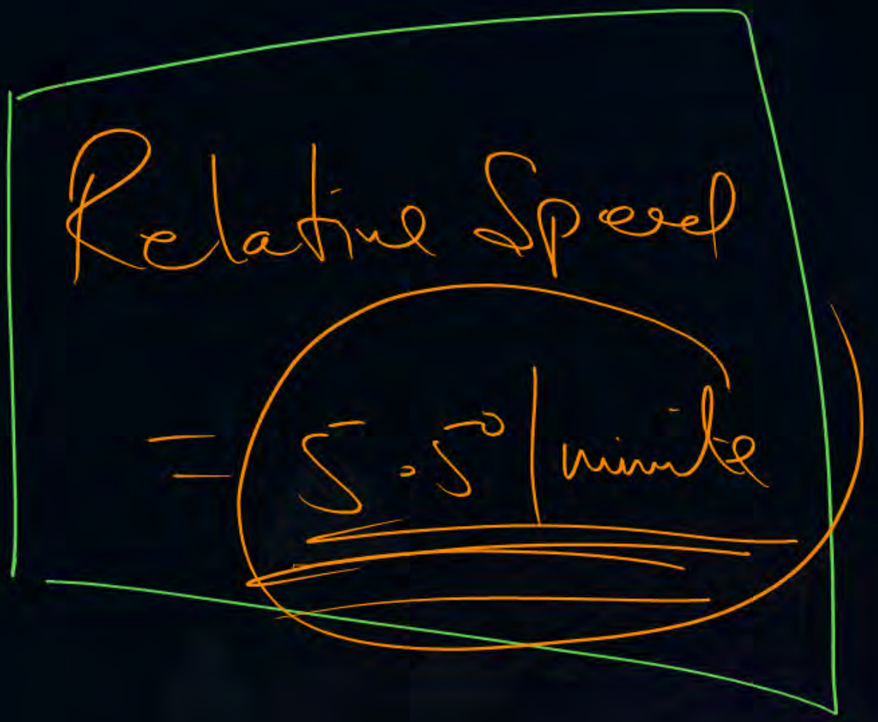
  6°/min.

- ✓ Hour Hand Covers 30° in 60 minutes 30°/60

  = ½ ° / min

  or

  0.5°/min



Different Patterns of Questions:

I ⇒ Time ⇒ Angle?

II ⇒ Angle ⇒ Time?

II ⇒ Gain OR loss

First Pattern:

7:00

210°

270°

9:00

150°

5:00

60°

2:00



Random time given:

4:45 → 127.5° ↺ 232.5°

3:14 → 13° ↻ 347°

4 ⇒ 120°

45 × 5.5 ⇒ 247.5°

247.5° − 120° = 127.5°

3 ⇒ 90°

14 × 5.5 ⇒ 77°

77° − 90° = 13°

Questionnaire:

5:30 → 15·0w

5 ⇒ 150°

30 × 5·5 ⇒ 165°

15°



Questionnaire:

3:15 ⇒ 7.5° ✓

352.5°

3 ⇒ 90°

15 × 5.5 ⇒ 82.5°

7.5



Questionnaire:

4:40 → 100° ∠

260° ∠

4 ⇒ 120°

40 × 5.5 ⇒ 220°

100°



[MCQ]

#Q. At 9:45, the two hands of a clock make an angle of?

Assignment

A 60°

B 45°

C 33 ⅓°

D 22 ½°



2 mins Summary

Topic Clock

Time → Angle?

5.5°/min

THANK - YOU